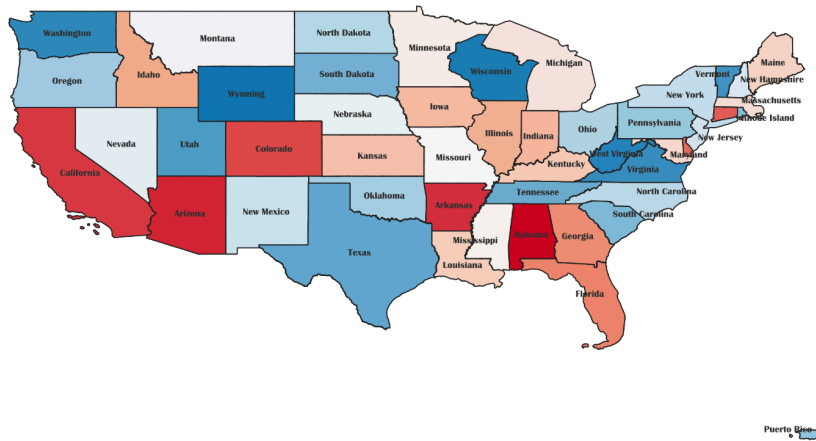


PS4 Assignment

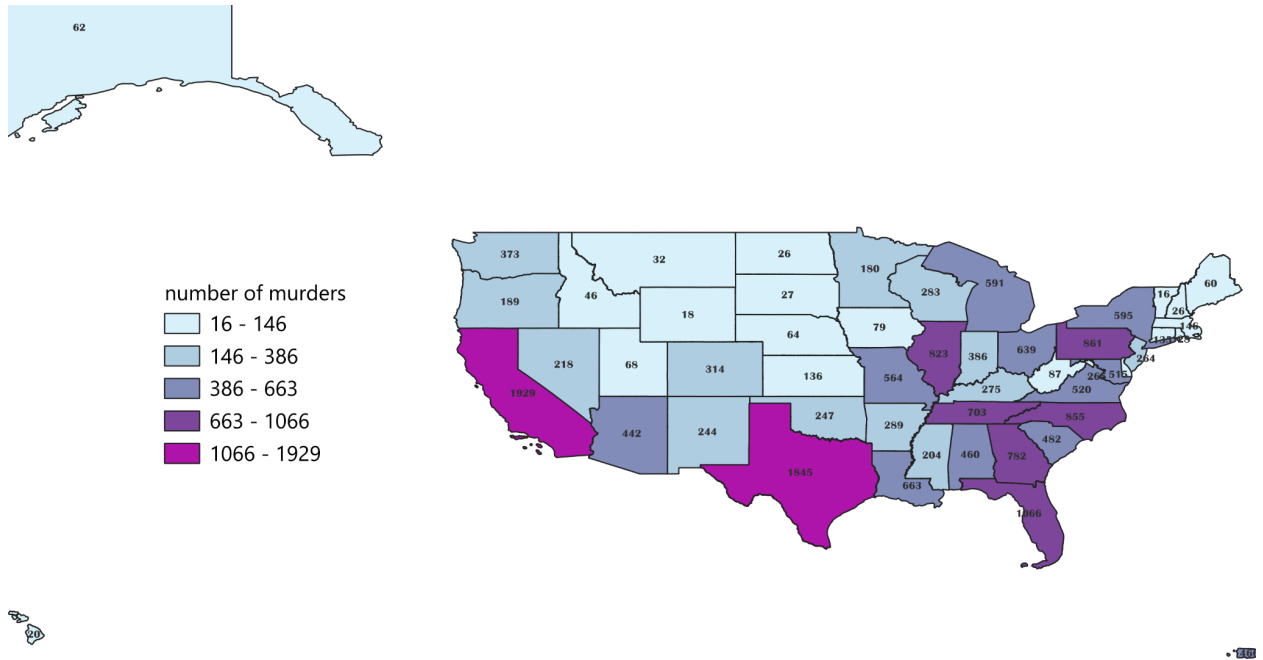
Map 1: Categorical Variable



Map Interpretation:

Since my research is focused on all 50 states in the United States of America, this map was created to illustrate the name variables of my study.

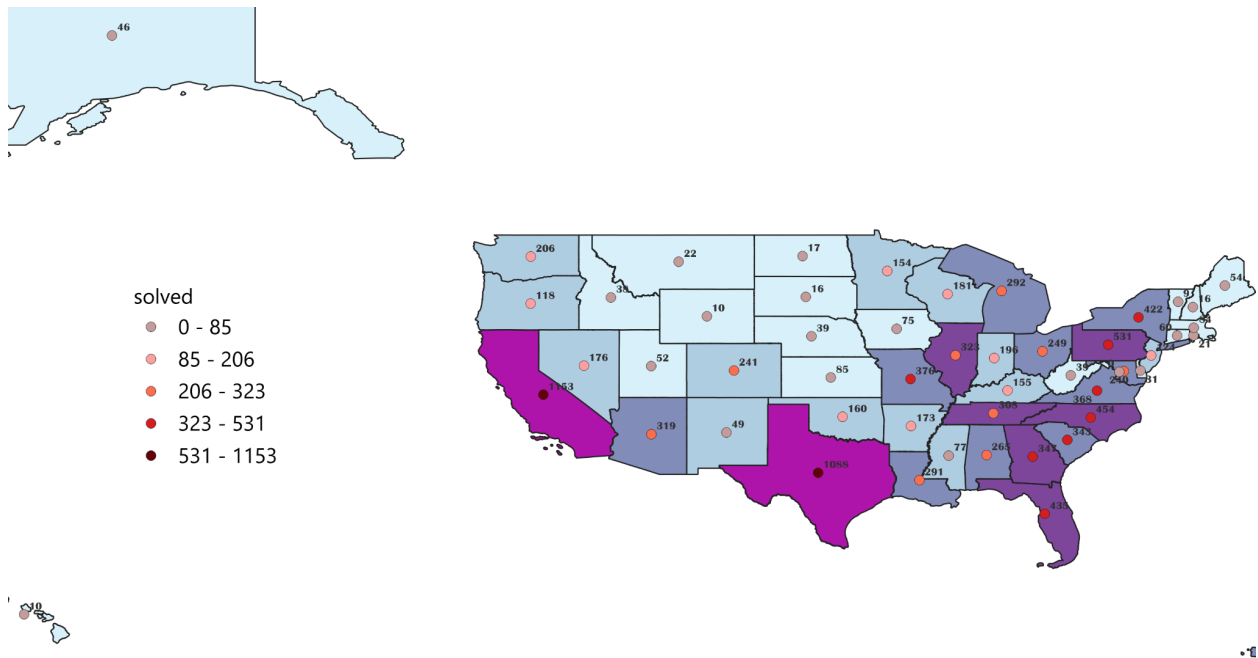
Map 2: Numerical Variable



Map Interpretation:

This map focuses on the number of murders reported in the year of 2023 in each state of the United States. The number of murders increases with brightness/darkness of color. For example, Wyoming has a pretty low number in comparison to California. Overall, we can see that the lower number of murders reported fall in the midwest while higher numbers of murders reported are along the east coast. California and Texas hold the top two spots in the largest number of murders reported.

Map 3: Two Variables



Map Interpretation:

This map focuses on the number of murders solved in the year of 2023 in each state of the United States. As one would suspect, the states with higher number of murders reported have a higher number of murders solved. I would like to dive into the top five leading states and explain the efficiency of each for the year of 2023. In 2023, California had around 600 different law enforcement agencies. Based on the provided data, California's overall efficiency of getting cases solved was approximately 60%. In 2023, Texas had around 2768 different law enforcement agencies. Based on the provided data, Texas's overall efficiency of getting cases solved was approximately 60%. In 2023, Florida had around 373 different law enforcement agencies. Based on the provided data, Florida's overall efficiency of getting cases solved was approximately 41%. In 2023, Pennsylvania had around 1300 different law enforcement agencies. Based on the provided data, Pennsylvania's overall efficiency of getting cases solved was approximately 62%. In 2023, North Carolina had around 504 different law enforcement agencies. Based on the provided data, North Carolina's overall efficiency of getting cases solved was approximately 53%. These show generally poor efficiency rates of law enforcement agencies solving cases.

Point Data Idea

Unfortunately, both of my datasets do not include population amounts for each of the states for the year of 2023, but I believe that variable could be extremely insightful for the discovery of efficiency and safety. I think that by possibly bringing in a population as a variable of analysis, the values for the number of murders reported and solved would make even more sense.